Department of Chemical Engineering, Chulalongkorn University (teaching)

MEA CO2 Capture Ver 1 · operator training simulator

CAUSE AND EFFECT MATRIX

SIS interlocks I-1 / I-2 / I-3 and DCS alarms

MEA CO2 capture unit: absorber T-101 / stripper T-102 (30 wt% MEA)

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Drawings	CC-PID-101 rev A · MEA-PFD-001 rev 0
Related	SRS MEA-IN-SRS-001 · HAZOP MEA-SA-HZP-001 rev A
Contents	Cover (A4) + sheet 1 of 1 (A3 landscape)

Revision record

REV	DATE	DESCRIPTION	ORIGINATOR	CHECKER	APPROVER
0	28-Oct-2026	Issued for teaching	Apinan Soottitantawat		

Teaching document for the MEA CO2 capture operator training simulator. Not for construction. Causes, settings and effects are read from the simulator model (src/model.js) and its alarm list; proposed rows come from the LOPA recommendations. Failure data and hazard limits are assumed (see the SRS). Checker and approver are left blank: this issue has not been checked or approved.

CAUSE AND EFFECT MATRIX · SIS INTERLOCKS I-1 / I-2 / I-3 AND DCS ALARMS · MEA-IN-CE-001 REV 0

CAUSES								EFFECTS									NOTES
ITEM	TAG	SERVICE	DIR.	SET	UNIT	DELAY s	INTERLOCK	K-101 flue gas blower	XV-101 flue-gas damper	XV-207 LP steam to E-102	FIC-101 (K-101 speed)	FIC-207 (LP steam)	P-101 rich pump	P-102 lean pump	First-out indication (BMS panel)	DCS alarm / annunciator	
		TRIP ACTION ▶						STOP	CLOSE	CLOSE	MAN 0 %	MAN 0 %	STOP	STOP	1ST OUT	ALARM	
SIS TRIPS (I-1 UNIT TRIP · I-2 REBOILER TRIP · I-3 PUMP STOPS)																	
001	**HS-ESD**	Emergency shutdown pushbutton (BMS panel)	—	—		0	I-1	X	X	X	X	X			X	X	Manual I-1. Latched: Reset, then permissives and Open.
002	**LSHH-109**	T-101 sump level high-high (own sensor LS-109)	HH	90	%	5	I-1	X	X	X	X	X			X	X	I-1, latched. Reset blocked while level > 88 %.
003	**PSHH-301**	T-102 / V-101 pressure high-high (own sensor PS-301)	HH	250	kPa a	3	I-2			X		X			X	X	I-2. Reset when P < 245 kPa a.
004	**TSHH-206**	Reboiler E-102 temperature high-high (own sensor TS-206)	HH	130	°C	5	I-2			X		X			X	X	I-2. Reset when T < 128 °C.
005	**LSLL-208**	T-102 sump level low-low (own sensor LS-208)	LL	10	%	5	I-2			X		X		X	X	X	I-2 steam shut + I-3 P-102 stop. Reset when level > 15 %.
006	**LSLL-109**	T-101 sump level low-low (own sensor LS-109)	LL	5	%	3	I-3						X			X	I-3: stops P-101 (no first-out). Restart P-101 by hand.
DCS ALARMS (ALM LIST · 5 s ON/OFF DELAY)																	
007	**FIC-101**	Flue gas to the absorber	L	150	kNm³/h	5	DCS									X	5 s on/off delay; suppressed without flue gas
008	**FIC-101**	Flue gas to the absorber	H	205	kNm³/h	5	DCS									X	5 s on/off delay; suppressed without flue gas
009	**AI-101**	CO2 in the flue gas	L	3.2	mol%	5	DCS									X	5 s on/off delay; suppressed without flue gas
010	**AI-101**	CO2 in the flue gas	H	5	mol%	5	DCS									X	5 s on/off delay; suppressed without flue gas
011	**AY-112**	CO2 capture low	L	88	%	5	DCS									X	5 s on/off delay; suppressed without flue gas
012	**AI-111**	CO2 in the treated gas high	H	0.55	mol%	5	DCS									X	5 s on/off delay; suppressed without flue gas
013	**PDI-102**	Absorber ΔP high	H	6	kPa	5	DCS									X	5 s on/off delay; suppressed without flue gas
014	**AY-102**	Absorber approaching flood	H	80	%	5	DCS									X	5 s on/off delay; suppressed without flue gas
015	**LIC-109**	Absorber sump level	L	30	%	5	DCS									X	5 s on/off delay
016	**LIC-109**	Absorber sump level	H	70	%	5	DCS									X	5 s on/off delay
017	**TIC-110**	Treated gas hot (water loss)	H	45	°C	5	DCS									X	5 s on/off delay; suppressed without flue gas
018	**FIC-201**	Lean solvent flow	L	220	t/h	5	DCS									X	5 s on/off delay; suppressed without lean circulation
019	**FIC-201**	Lean solvent flow	H	330	t/h	5	DCS									X	5 s on/off delay; suppressed without lean circulation
020	**TIC-202**	Lean solvent to T-101 hot	H	45	°C	5	DCS									X	5 s on/off delay; suppressed without lean circulation
021	**AI-203**	Lean loading high	H	0.28	mol/mol	5	DCS									X	5 s on/off delay; suppressed without lean circulation
022	**AI-113**	Rich loading high	H	0.485	mol/mol	5	DCS									X	5 s on/off delay; suppressed when not capturing
023	**TIC-206**	Reboiler temperature	L	117	°C	5	DCS									X	5 s on/off delay; suppressed with the steam off
024	**TIC-206**	Reboiler temperature	H	123	°C	5	DCS									X	5 s on/off delay; suppressed with the steam off
025	**LIC-208**	Stripper sump level	L	30	%	5	DCS									X	5 s on/off delay
026	**LIC-208**	Stripper sump level	H	70	%	5	DCS									X	5 s on/off delay
027	**AY-210**	Stripper approaching flood	H	80	%	5	DCS									X	5 s on/off delay; suppressed with the steam off
028	**PIC-301**	Stripper / reflux drum pressure	L	165	kPa a	5	DCS									X	5 s on/off delay; suppressed with the steam off
029	**PIC-301**	Stripper / reflux drum pressure	H	190	kPa a	5	DCS									X	5 s on/off delay; suppressed with the steam off
030	**TIC-302**	Condenser outlet hot	H	50	°C	5	DCS									X	5 s on/off delay; suppressed with the steam off
031	**LIC-303**	Reflux drum level	L	25	%	5	DCS									X	5 s on/off delay
032	**LIC-303**	Reflux drum level	H	75	%	5	DCS									X	5 s on/off delay
033	**AY-212**	Specific reboiler duty high	H	4.3	GJ/t	5	DCS									X	5 s on/off delay; suppressed when not capturing
034	**AIT-213**	MEA concentration	L	28	wt%	5	DCS									X	5 s on/off delay
035	**AIT-213**	MEA concentration	H	32	wt%	5	DCS									X	5 s on/off delay
PROPOSED SIFs FROM THE LOPA RECOMMENDATIONS (NOT IN THE SIMULATOR)																	
036	**PSHH-301**	*Proposed SIF-2R: PSHH-301 also stops P-101*	*HH*	*250*	*kPa a*	*3*	*I-2 ext.*			P		P	P		P	P	*PROPOSED (LOPA LR2, HAZOP R29). Not in the simulator.*
037	**EUV-MCC**	*Proposed SIF-7R: MCC bus under-voltage (loss of power)*	*L*	—		*1*	*I-2 new*			P		P			P	P	*PROPOSED (LOPA LR11, HAZID HD-29). Delay 1 s ride-through assumed.*
038	**LSHH-208**	*Proposed SIF-6R: T-102 sump level high-high (new sensor LS-208H)*	*HH*	*85*	*%*	*3*	*new*						P		P	P	*PROPOSED (LOPA LR8, HAZOP R18). Also shuts a new XV-208 on the make-up water. Delay 3 s assumed.*
039	*SPARE*																
040	*SPARE*																

NOTES

1. Trip switches have their own sensors (LS-109, LS-208, TS-206, PS-301), independent of the control transmitters LT-109, LT-208, TT-206 and PT-301. The model trips on the true process value.
2. Delay = time the cause must persist before the effects act (sisStep in src/model.js). I-1 latches the unit TRIPPED; I-2 latches XV-207 shut until Reset.
3. On I-1 the solvent keeps circulating (P-101 and P-102 are not stopped) so that both columns stay wetted and the hot reboiler cools down.
4. FIC-101 and FIC-207 go to MAN at 0 % so that Reset and Open cannot restart the blower or re-open the steam at full output.
5. Alarm rows are DCS alarms from the ALM list: 5 s on/off delay (ISA-18.2 practice), suppressed while their section is out of service.
6. X = acts in the simulator. P = proposed effect of a LOPA recommendation, not in the simulator and not checked against it.
7. XV-101 and XV-207 close on de-energise (spring return) and the motors stop on de-energise of the MCC interposing relay: fail-safe, assumed.
8. Effects were read from bmsTrip() and sisStep() by running the model (tabs/srs/tools/effects.js); part 5 of the C&E tab repeats that test in the page.

LEGEND

X	SIS cause acts on this effect (in the simulator)
X	DCS alarm limit (alarm only)
P	Proposed by the LOPA, not in the simulator
TAG	Trip tag (SIS cause)
HH/LL	Trip high-high / low-low; H/L alarm high / low

I-1 unit trip: K-101 stop, XV-101 and XV-207 close, FIC-101 and FIC-207 MAN 0 %, latched.

I-2 reboiler trip: XV-207 close, FIC-207 MAN 0 %.

I-3: LSLL-109 stops P-101; LSLL-208 stops P-102.

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